



Entergy Nuclear Operations, Inc.
Pilgrim Nuclear Power Station
600 Rocky Hill Road
Plymouth, MA 02360

Peter J. Miner
Manager, Regulatory Assurance

10 CFR 50.73

2.19.014

March 4, 2019

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Licensee Event Report 2019-002-00, Failure of Main Steam Isolation Valve
Limit Switch Results in a Condition Prohibited by Technical Specifications

Pilgrim Nuclear Power Station
NRC Docket No. 50-293
Renewed Facility Operating License No. DPR-35

The enclosed Licensee Event Report 2019-002-00, Failure of Main Steam Isolation Valve
Limit Switch Results in a Condition Prohibited by Technical Specifications, is submitted in
accordance with Title 10 Code of Federal Regulations 50.73.

There are no regulatory commitments contained in this letter.

If you have any questions or require additional information, please contact me at
508-830-7127.

Sincerely,

A handwritten signature in black ink, appearing to read "Peter J. Miner".

Peter J. Miner

PJM/rb

Enclosure: Licensee Event Report 2019-002-00, Failure of Main Steam Isolation Valve
Limit Switch Results in a Condition Prohibited by Technical Specifications

IE22
NRR

cc: NRC Region I Regional Administrator
NRC Senior Resident Inspector - Pilgrim
NRC NRR Project Manager - Pilgrim

Enclosure

2.19.014

Licensee Event Report 2019-002-00, Failure of Main Steam Isolation Valve Limit Switch Results in a
Condition Prohibited by Technical Specifications

8NRC FORM 366 (04-2017)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104		EXPIRES: 03/31/2020					
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)												
1. FACILITY NAME Pilgrim Nuclear Power Station					2. DOCKET NUMBER 05000-293			3. PAGE 1 OF 4				
4. TITLE Failure of Main Steam Isolation Valve Limit Switch Results in a Condition Prohibited by Technical Specifications												
5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED			
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER		
01	03	2019	2019	- 002	- 00	03	04	2019	N/A	N/A		
									FACILITY NAME	DOCKET NUMBER		
									N/A	N/A		
9. OPERATING MODE		11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)										
N		☐ 20.2201(b)			☐ 20.2203(a)(3)(i)			☐ 50.73(a)(2)(ii)(A)			☐ 50.73(a)(2)(vii)(A)	
		☐ 20.2201(d)			☐ 20.2203(a)(3)(ii)			☐ 50.73(a)(2)(ii)(B)			☐ 50.73(a)(2)(viii)(B)	
		☐ 20.2203(a)(1)			☐ 20.2203(a)(4)			☐ 50.73(a)(2)(iii)			☐ 50.73(a)(2)(ix)(A)	
		☐ 20.2203(a)(2)(i)			☐ 50.36(c)(1)(i)(A)			☐ 50.73(a)(2)(iv)(A)			☐ 50.73(a)(2)(x)	
10. POWER LEVEL 35		☐ 20.2203(a)(2)(ii)			☐ 50.36(c)(1)(ii)(A)			☐ 50.73(a)(2)(v)(A)			☐ 73.71(a)(4)	
		☐ 20.2203(a)(2)(iii)			☐ 50.36(c)(2)			☐ 50.73(a)(2)(v)(B)			☐ 73.71(a)(5)	
		☐ 20.2203(a)(2)(iv)			☐ 50.46(a)(3)(ii)			☐ 50.73(a)(2)(v)(C)			☐ 73.77(a)(1)	
		☐ 20.2203(a)(2)(v)			☐ 50.73(a)(2)(i)(A)			☐ 50.73(a)(2)(v)(D)			☐ 73.77(a)(2)(i)	
		☐ 20.2203(a)(2)(vi)			☒ 50.73(a)(2)(i)(B)			☐ 50.73(a)(2)(vii)			☐ 73.77(a)(2)(ii)	
			☐ 50.73(a)(2)(i)(C)			☐ OTHER			Specify in Abstract below or in NRC Form 366A			
12. LICENSEE CONTACT FOR THIS LER												
LICENSEE CONTACT Mr. Peter J. Miner - Regulatory Assurance Manager									TELEPHONE NUMBER (Include Area Code) 508-830-7127			
13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT												
CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX			
X	SB	33	N007	Y								
14. SUPPLEMENTAL REPORT EXPECTED						15. EXPECTED SUBMISSION DATE		MONTH	DAY	YEAR		
☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO												
ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines) On January 5, 2019 during the performance of Main Steam Isolation Valve (MSIV) surveillance testing, control room alarm MSIV Not Full Open did not annunciate when MSIV AO-203-1C was fully closed for stroke time testing. The MSIV Not Full Open alarm is expected when any MSIV position RPS limit switch opens. The cause was traced to failure of the MSIV position Reactor Protection System (RPS) limit switch LS6 to properly open and the associated logic system channel was placed in the tripped condition. On February 7, 2019, a detailed review of plant computer and annunciator system alarm data discovered that the MSIV position RPS limit switch LS6 did not open as expected when MSIV AO-203-1C was closed on January 3, 2019 and the associated logic system channel was not placed in a tripped condition within the time constraints specified in Technical Specifications. The allowable action time to trip (de-energize) the RPS channel associated with failed limit switch LS6 on MSIV AO-203-1C was exceeded because operations personnel did not correctly interpret the absence of the MSIV Not Full Open alarm upon closure of the MSIV on January 3, 2019. At the time of the event, the reactor mode selector switch was in the RUN position and the reactor was at 35 percent reactor thermal power. This event had no impact on the health and/or safety of the public. This report is submitted in accordance with Title 10 Code of Federal Regulations 50.73(a)(2)(i)(B); Operation or Condition Prohibited by Technical Specifications.												

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
Pilgrim Nuclear Power Station		05000- 293	YEAR	SEQUENTIAL NUMBER	REV NO.
			2019	- 002	- 00

NARRATIVE**BACKGROUND**

The Reactor Protection System (RPS) provides timely protection against the onset and consequences of conditions that threaten the integrity of the fuel clad and reactor coolant pressure boundary.

The Main Steam System conducts steam from the Reactor Pressure Vessel (RPV), via four steam lines, through the Primary Containment to the Turbine. Each steam line has two, normally open, Main Steam Isolation Valves (MSIVs), one inside and one outside the Primary Containment. The MSIVs close automatically upon receipt of certain isolation signals.

The RPS MSIV closure signals are initiated from limit switches (valve position) located on each of the eight MSIVs. Each MSIV has two limit switches; one inputs to RPS trip system A while the other inputs to RPS trip system B. Each RPS trip system receives an input from four MSIV-closure channels, each consisting of two limit switches (one for the inboard MSIV and one for the outboard MSIV in the same steam line) in series with a sensor relay. The design permits closure of any two lines without a full scram being initiated. There are four limit switch contacts on each MSIV that provide valve stem position input for control room indication and inputs into the RPS. Limit switches 5 and 6 are closed only when the MSIV is fully open (>90 percent open) and supply the RPS circuitry.

It should be noted that prior to testing performed on January 5, 2019, there was an earlier failure of RPS limit switch LS5 on MSIV AO-203-1C that was discovered on November 26, 2018 during testing. Upon discovery of the LS5 failure, the associated logic system channel was placed in a tripped condition. After discovering the failure of LS6 on MSIV AO-203-1C on January 5, 2019, and placing the associated logic system channel in the tripped condition, both RPS MSIV closure channels associated with the "C" steamline MSIVs are in the tripped condition.

EVENT DESCRIPTION

On January 5, 2019 during the performance of MSIV surveillance testing, control room alarm MSIV Not Full Open did not annunciate when MSIV AO-203-1C was fully closed for stroke time testing. The MSIV Not Full Open alarm is expected when any MSIV position RPS limit switch opens. The cause was traced to failure of the MSIV position RPS limit switch LS6 to properly open and the associated logic system channel was placed in the tripped condition. On February 7, 2019, a detailed review of plant computer and annunciator system alarm data discovered that the MSIV position RPS limit switch LS6 did not open as expected when MSIV AO-203-1C was closed on January 3, 2019 at 0917 hours and the associated logic system channel was not placed in a tripped condition within the time constraints specified in Technical Specifications.

The January 3, 2019 event occurred during power operation while at 35 percent reactor power. The reactor mode selector switch was in the RUN position.

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			2019	- 002	- 00

CAUSE OF THE EVENT

On January 3, 2019, both "C" steamline MSIVs were closed during normal low power operations as directed by operating procedure. RPS MSIV closure logic system testing was not being performed. Upon closure of MSIV AO-203-1C the MSIV Not Full Open alarm did not annunciate. The allowable action time to trip (de-energize) the RPS channel associated with failed limit switch LS6 on MSIV AO-203-1C was exceeded because operations personnel did not correctly interpret the absence of the MSIV Not Full Open alarm upon closure of the MSIV on January 3, 2019.

CORRECTIVE ACTIONS

MSIV AO-203-1C RPS limit switches are located in the Primary Containment and are not accessible during plant operations. It is our intention to initiate troubleshooting of the condition should an outage of sufficient duration and scope occur.

Any further corrective actions will be documented in the corrective action program.

SAFETY CONSEQUENCES

There were no actual consequences to safety of the general public, nuclear safety, industrial safety, or radiological safety for this event since the MSIV closure scram function within the RPS is designed to accommodate single failure within the logic system and therefore was capable of performing its intended safety function during this event.

REPORTABILITY

The NRC Operations Center was notified of the condition discovered on January 5, 2019 in accordance with Title 10 Code of Federal Regulations (CFR) 50.72 (b)(3)(v) "Any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to: (A) Shut down the reactor and maintain it in a safe shutdown condition" at 1730 hours via Event Number 53815. This 10 CFR 50.72 notification was subsequently retracted on February 11, 2019 based on an Engineering Evaluation that concluded that the MSIV position RPS logic was capable of performing its intended safety function.

Subsequent evaluation determined that the failure of limit switch LS6 on MSIV AO-203-1C is reportable under 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by Technical Specifications since the Technical Specification Required Action of Note 1 for Table 3.1.1, Condition a. to place the affected channel in a tripped condition, did not meet the Completion Time of 12 hours.

PREVIOUS EVENTS

There have been no events reported in the last three years related to failure of MSIV limit switches.

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REFERENCES

Condition Report CR-PNP-2019-00084

Condition Report CR-PNP-2019-00090

Condition Report CR-PNP-2019-00838